Improved Maximum Frequency Estimation With Application to Instantaneous Mean Frequency Estimation of Surface Electromyography

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***Abstract*—The purpose of this study was to improve the maximum-frequency estimation. Three methods to estimate the maximum frequency of a bandlimited signal with additive white noise were compared. Two existing methods, the threshold-crossing method (TCM) and the hybrid method, were modified for time-frequency representations. A novel approach, the running-block threshold method (RBTM), was introduced. Based on calculation of detection probability (sensitivity) the RBTM improved the maximum-frequency estimate as compared with the TCM. The maximum-frequency estimation methods were also used to determine the integration interval for instantaneous mean-frequency (IMNF) estimation from synthesized surface electromyography containing white noise. Results showed that the IMNF estimate was improved by using any of the three methods and that the RBTM gave the best IMNF estimate.**

***Index Terms*—Electromyography (EMG), instantaneous mean frequency, maximum frequency, varying signal-to-noise ratio.**

I. Introduction

SURFACE electromyography (EMG) may be modeled as a superposition of action potential trains of the recruited motor units [3]. The shapes of the action potentials, their firing pattern and conduction velocities will affect the frequency spectrum. Therefore, the analysis of frequency spectra of EMG signals has been widely used to characterize muscle fatigue and has attracted great interest in the fields of rehabilitation medicine, ergonomics, and sports medicine [9], [10]. The mean frequency (MNF) is commonly used as a spectral change indicator, but higher order spectral moments [14], [17] have also been used as spectral change indicators. Moreover, the median frequency has been proposed because it is not as sensitive to noise as the MNF [3]. However, the median frequency has a higher variance of the estimate for high signal-to-noise ratio (SNR) [11].

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During a static contraction the surface EMG signal is often assumed to be stationary and the Fourier transform is applied. Unfortunately, pure isometric contractions are not common in daily activities. Tasks similar to daily activities contain dynamic contractions, and therefore such types of contractions are desirable to study. However, during dynamic contractions, the stationarity assumption can be questioned and therefore in recent studies time-frequency representations (TFRs) [4], [7], [8], [12], [14], [24], and instantaneous mean frequencies (IMNFs) [12], [13], [15] have been used.

One critical problem when analyzing dynamic contractions is the dependence of the IMNF estimate on the SNR. During dynamic contractions it is sometimes unavoidable to have periods with low SNR, e.g., in ergonomic applications. An approach to reduce this effect, where the upper integration limit used in IMNF calculation was estimated, has been used [5], [15]. The estimated integration limit, called maximum frequency, was used when estimating the instantaneous mean and median frequency. The maximum frequency was estimated with the threshold crossing method (TCM) proposed by D'Alessio [6]. However, the extension of the TCM to TFRs was not derived theoretically.

This study introduces a new method, the running-block threshold method (RBTM), to estimate the maximum frequency of a bandlimited signal with additive white noise that can be used to estimate the MNF or IMNF from surface EMG signals with low and varying SNR. This new method is compared with two existing methods, the TCM and the hybrid method [18], which have been modified for TFRs.

II. Signal and Spectra Characteristics

A surface EMG signal $s(t)$ can be modeled as a bandlimited zero mean Gaussian process [3] and the noise $n(t)$ that is superimposed on the signal can be assumed to be a white Gaussian process. The recorded signal $x(t)$ can be described as

$$x(t) = s(t) + n(t) \tag{1}$$

where the signal and noise are assumed to be independent. The noise power density N_0 can often be estimated from $x(t)$ when the signal $s(t)$ is absent or from the high-frequency contents of the signal $x(t)$. The EMG signal's power density $\mathcal{P}_s$ cannot be directly estimated because the noise is always present. However,

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by estimating the total power density $\mathcal{P}$ and the noise power density, an estimate of $\mathcal{P}_s$ and SNR can be obtained from

$$\mathcal{P} = \mathcal{P}_s + N_0 \text{ and } \mathrm{SNR} = 10\log_{10}\left(\frac{\mathcal{P}_s}{N_0}\right). \tag{2}$$

When the signal consists only of white Gaussian noise with unit variance, the coefficients of a nonnegative bilinear TFR are approximately chi-squared distributed [19]. Thus, the distribution of the spectral bins can be described with a gamma distribution with the scale parameter $\lambda = 1/2$ and the shape parameter α equal to half the number of degrees of freedom. Its probability density function is

$$g(x|\alpha,\lambda) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x} \tag{3}$$

where $\Gamma(\alpha) = \int_0^\infty u^{\alpha-1}e^{-u}du$, especially

$$\alpha \in \mathbb{Z}^+ \Rightarrow \Gamma(\alpha) = (\alpha-1)! \tag{4}$$

When the noise does not have unit variance, the scale parameter λ will change, so by fitting a gamma distribution the variance will also be taken into consideration. For a nonnegative bilinear transform, when the coefficients of the transform have two degrees of freedom, for example in a periodogram, the distribution simplifies to an exponential distribution.

When an EMG signal is present, the frequency bins will no longer be identically distributed and therefore a local distribution, in the time-frequency plane, will be used instead

$$g_L(x|\alpha,\Lambda_L) = \frac{\Lambda_L^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\Lambda_L x} \tag{5}$$

$$\Lambda_L = \frac{\lambda}{1+10^{\mathrm{SNR}_L/10}} \triangleq C_L\lambda \tag{6}$$

where SNR_L is the local SNR.

III. Maximum Frequency Estimation

For most signals, as in the case of surface EMG signals, there are no well-defined maximum frequencies. Therefore, the maximum frequency obtained from the methods in this study cannot be compared with a true maximum frequency. Instead, we can calculate the detection probability (sensitivity), which is defined as the probability of detecting frequency bins that are derived from the signal.

Traditionally, MNF has been estimated from the power spectrum of the signal, which is obtained with a Fourier transform. For a nonstationary signal the frequency content changes with time, and it would be better to have an MNF that is dependent on the time, i.e., IMNF. If a TFR, which may be thought as a frequency spectrum as a function of time, is calculated, then the IMNF can be estimated according to

$$\mathrm{IMNF}(t) = \frac{\int_{f_L}^{f_H} f \cdot \mathrm{TFR}(f,t)df}{\int_{f_L}^{f_H} \mathrm{TFR}(f,t)df} \tag{7}$$

where $[f_L, f_H]$ is the bandwidth on which the TFR has been calculated. It is known that the MNF and IMNF estimates are affected by the high-frequency noise and that the surface EMG signal is bandlimited. Therefore, it was suggested that the IMNF should be calculated from f_L to a maximum frequency dependent on time t, here referred to as $f_{\max}(t)$, where there are significant EMG signal components compared to the noise [5], [15].

A. Threshold Crossing Method

The TCM has been used to find maximum frequencies of Doppler signals [6], [16], [18], [22] and EMG signals [5], [15]. The TCM finds the maximum frequency from a power spectrum by setting a threshold, and when at least r out of m frequency bins exceed the threshold, starting from the high frequency end of the spectrum, then the highest of those frequency bins are chosen as $f_{\max}$. The threshold is calculated by specifying the specificity, P_S. The specificity is here defined as the probability that a spectral bin can correctly be recognized as noise. If P_S, r, and m are specified, the probability P_e that a spectral bin will exceed the threshold can be calculated by using the binomial distribution, under the assumption that the spectral bins are independent. That is

$$P_S = 1 - \sum_{j=r}^{m} \binom{m}{j} P_e^j (1-P_e)^{m-j}. \tag{8}$$

If the spectral bins are exponentially distributed with parameter λ, then

$$P_e = \int_{T_h}^{\infty} \lambda e^{-\lambda t}dt = e^{-\lambda T_h} \tag{9}$$

and we get the threshold T_h by

$$T_h = -\frac{\ln P_e}{\lambda}. \tag{10}$$

If m is small, the m spectral bins could be assumed to have the same local probability distribution and the detection probability, P_D, can be calculated according to

$$P_D = \sum_{j=r}^{m} \binom{m}{j} \left(P_e^{C_L}\right)^j \left(1-P_e^{C_L}\right)^{(m-j)}. \tag{11}$$

Especially when $r = m$, we have

$$r = m \Rightarrow P_e = (1-P_S)^{1/r} \Rightarrow P_D = P_e^{rC_L} = (1-P_S)^{C_L} \tag{12}$$

which implies that P_D is independent of r and m when they are equal and the spectral bins are exponentially distributed.

If the spectral bins are gamma distributed, then the threshold T_h should instead be computed as

$$T_h = G^{-1}(1-P_e|\alpha,\lambda) \tag{13}$$

where G is the cumulative distribution function

$$G(x|\alpha,\lambda) = \frac{\lambda^\alpha}{\Gamma(\alpha)} \int_0^x t^{\alpha-1}e^{-\lambda t}dt. \tag{14}$$

The detection probability will then be

$$P_D = \sum_{j=r}^{m} \binom{m}{j} (1-G(T_h|\alpha,C_L\lambda))^j \cdot (G(T_h|\alpha,C_L\lambda))^{(m-j)}. \tag{15}$$

From the computational and practical point of view, we can simplify the above formula by using the standard gamma distribution (denoted by G_v) having only the shape parameter and the following relationship:

$$G_V(x|\alpha) = G(x|\alpha, 1). \tag{16}$$

It can be proven (see Appendix for the proof) that

$$G\left(G^{-1}(P_S|\alpha,\lambda)|\alpha, C_L\lambda\right) = G_V\left(C_L G_V^{-1}(P_S|\alpha)|\alpha\right) \tag{17}$$

which gives us the following alternative calculation of the detection probability

$$P_D = \sum_{j=r}^{m} \binom{m}{j} \left(1 - G_V\left(C_L G_V^{-1}\left(P_S|\alpha\right)|\alpha\right)\right)^j \cdot \left(G_V\left(C_L G_V^{-1}\left(P_S|\alpha\right)|\alpha\right)\right)^{(m-j)}. \tag{18}$$

A TFR can be seen as a power spectrum for each time sample, and by applying the TCM on each of those power spectra, the maximum frequency as a function of time will be obtained. The method cannot be directly applied if the frequency bins are dependent, because we made an assumption of independence in the calculations. However, instead of comparing adjacent frequency bins, if the frequency bins located with an offset from each other corresponding to the dependency in the representation are used, then the frequency bins could be assumed to be independent and the TCM could be applied. The dependency could be calculated or estimated from the autocorrelation of the spectra from the noise.

B. Hybrid Method

The hybrid method that has been used to find maximum frequencies of Doppler signals [16], [18], [22] finds $f_{\max}$ as the intersection between a straight line and the integrated power spectrum [18]. The line, which starts at the high frequency end of the integrated power spectrum, has a slope that is dependent on the noise power density. For a power spectrum ψ, the maximum frequency is

$$f_{\max} = \min\left\{ f^*: \int_{f_L}^{f^*} \psi(f)df = \int_{f_L}^{f_H} \psi(f)df - K*N_0\left(f_H - f^*\right) \right\} \tag{19}$$

where K is a parameter that has to be specified and the value of K should be greater or equal to one. The hybrid method could easily be adapted to a time-dependent power spectrum by simply making $f_{\max}$ time dependent, that is, for each t

$$f_{\max}(t) = \min\left\{ f^*(t): \int_{f_L}^{f^*(t)} \mathrm{TFR}(f,t)df = \int_{f_L}^{f_H} \mathrm{TFR}(f,t)df - K*N_0\left(f_H - f^*(t)\right) \right\}. \tag{20}$$

C. Running Block Threshold Method

The proposed method finds the maximum frequency, beginning from the high frequency end of the spectrum, by adding n frequency bins together and comparing them with a threshold. When the sum of the frequency bins exceeds the threshold, the highest of those frequency bins is chosen as $f_{\max}$. For a power spectrum ψ, the maximum frequency $f_{\max}$ that corresponds to the maximum-frequency bin k is defined as

$$f_{\max} = \max\left\{ k: \sum_{j=0}^{n-1} \psi\left(k - R*j\right) > T \right\} \tag{21}$$

where T is a threshold. If there is a redundancy in the representation, the n frequency bins should be located with the offset R from each other that corresponds to the dependency in the representation so that the frequency bins are independent.

The sum of n independent gamma distributed random variables with parameters α and λ is still distributed as a gamma distribution, but with parameters $n\alpha$ and λ.

Given the specificity P_S of the method, the threshold T can be calculated as follows:

$$T = G^{-1}(P_S|n\alpha, \lambda). \tag{22}$$

If n is small, the n spectral bins could be assumed to have the same local probability distribution and the detection probability of the RBTM can be obtained from

$$P_D = 1 - G(T|n\alpha, \Lambda_L) \tag{23}$$

or from

$$P_D = 1 - G_V\left(C_L G_V^{-1}(P_S|n\alpha)|n\alpha\right). \tag{24}$$

Equation (24) can be obtained from (17), (22), and (23).

For a TFR, the maximum frequency for a time-instant t that corresponds to the maximum-frequency bin k is defined as

$$f_{\max}(t) = \max\left\{ k(t): \sum_{j=0}^{n-1} \mathrm{TFR}\left(k(t) - R*j, t\right) > G^{-1}(P_S|n\alpha, \lambda) \right\} \tag{25}$$

where R is the offset that takes the redundancy in the representation into consideration.

IV. Simulation Procedure

To compare the different methods when used to determine the integration interval, signals were synthesized by filtering white noise with an approximation of the Shwedyk model [21] of the surface EMG signal spectrum. The signal spectrum [23] was given by

$$\psi(f) = \frac{c f_h^4 f^2}{\left(f^2 + f_l^2\right)\left(f^2 + f_h^2\right)^2}. \tag{26}$$

The cutoff frequencies were linearly changed from 60 and 120 Hz down to 30 and 60 Hz, with an assumed sampling rate

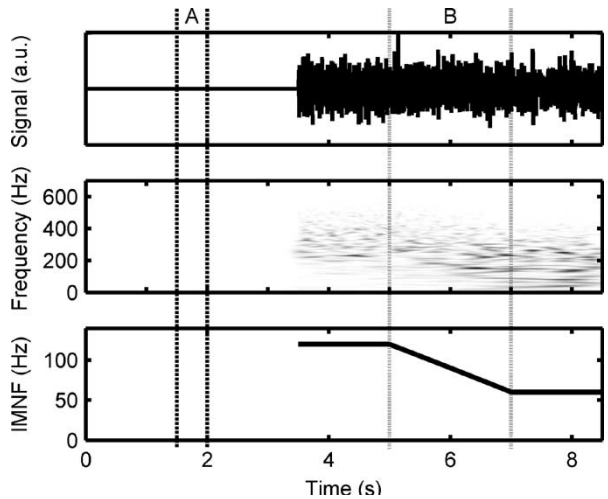

Fig. 1. Simulated signal without added noise, the estimated TFR and the theoretical IMNF. From the interval A of the signal, the noise power density is estimated. The interval B is the nonstationary part of the simulated signal and from this interval the IMNF is estimated and the rms error is calculated.

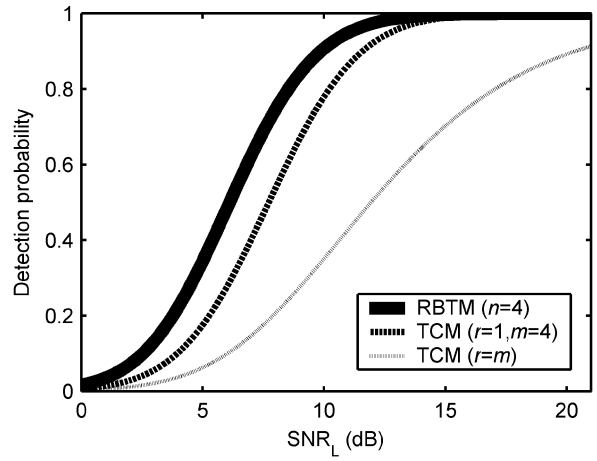

Fig. 2. The detection probability at a 99.999% specificity for the threshold crossing method and the running block threshold method for different local SNRs.

of 2 kHz. Note that c was used to normalize the output of the filter so that the signal energy was independent of the cutoff frequencies. The filter coefficients for a 25-order infinite-impulse response (IIR) filter were obtained as a least-square fit to the frequency response. The filter coefficients were modified for every time-sample to get a linearly changing MNF.

To the signals, which were 2 s long, additional stationary intervals were generated before and after the nonstationary interval. The cutoff frequencies of the initial and final stationary intervals were equal to the initial and final part of the nonstationary interval, respectively. Finally, a silent period was added at the beginning and white Gaussian noise was added to the whole signal. All statistics, except the noise power density, were estimated from the nonstationary part of the signal after the noise was added. The noise power density was estimated from half a second of the silent period with the added noise. The stationary parts were added to avoid window effects when calculating the TFR. An example of a signal without the added noise can be seen in Fig. 1.

For each signal a TFR using the continuous wavelet transform (CWT) was calculated with 128 frequencies from 15 to 700 Hz. Although the CWT originally was a time-scale representation, there are strong connections with time-frequency representations [20]. The TFR was calculated for different SNRs, obtained by adding different levels of white Gaussian noise to the signals. SNR values ranging from 0 to 21 dB in step of 3 dB were used. The CWT was calculated with a complex Morlet wavelet with length 64 at coarsest scale, and then used to estimate the IMNF with the different methods of estimating the maximum frequency. The Morlet wavelet was selected because it gives good resolution in both time and frequency. For each signal the residual errors with reference to the true IMNF ($\mathrm{IMNF}_{\mathrm{true}}$) calculated from the theoretical frequency response were calculated. The root mean square (rms) was calculated for the residual error e of the IMNF using the different methods to determine the integration interval, where

$$e(t) = \mathrm{IMNF}(t)_{\mathrm{estimated}} - \mathrm{IMNF}(t)_{\mathrm{true}}. \tag{27}$$

A low specificity of the methods will produce a low bias, but a high variance. If there is a low probability that the methods are correct, they can discover the signal earlier but will instead sometimes make big mistakes and it will result in a low bias but a high variance. A high specificity will have the opposite effect. Because the tradeoff between bias and variance could be to some extent controlled with the specificity of the TCM and the RBTM, the rms of the error was chosen for comparison. The simulation was done for 1000 signals for every SNR value.

According to the original article on TCM [6], a typical value for P_S could be 99%. However, choosing a low specificity will produce a high variance of the estimated maximum frequency, and that will be seen as spikes in the estimated IMNF. Therefore, the specificity was chosen higher, 99.999%, for the TCM and the RBTM. A more stable IMNF estimate does not need to be filtered, and a better comparison with the hybrid method was obtained.

Two sets of parameter values ($r = m = 2$ and $r = 1, m = 4$) were chosen for the TCM and $n = 4$ was chosen for the RBTM. The offset R used in the RBTM and the TCM were estimated from the autocorrelation of the noise spectra. When the autocorrelation was lower than 0.1 the frequency bins were assumed to be independent. For the hybrid method, different parameter values were tested and $K = 2$ was chosen as an appropriate value for the transform used.

All implementations were done in MATLAB® 6.5 with the Signal Processing toolbox 6.0, Statistics toolbox 4.0 (The MathWorks Inc., Natick, MA) and the Time-Frequency toolbox [1].

V. Results

The detection probability of exponentially distributed spectral bins was clearly higher for the RBTM than for the TCM (Fig. 2). For the RBTM, the detection probability increased when choosing a higher value for the parameter n. However, a large n gave an uncertainty of the location of $f_{\max}$ and therefore restrained to $n = 4$. The detection probability of exponentially distributed spectral bins was for the TCM found to be highest, with condition $m \leq 4$, when the parameter values were chosen as $r = 1$ and $m = 4$.

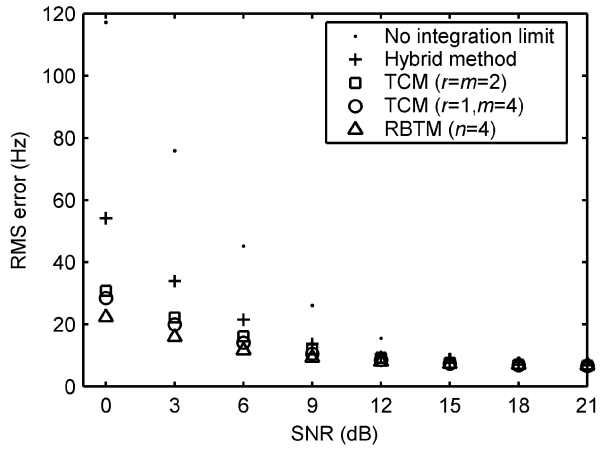

Fig. 3. Results from simulation with different methods to estimate the maximum frequency, which are used to calculate the IMNF from a CWT. The rms of the residual error of the estimated IMNF has been calculated and the mean of the 1000 simulations is shown for different SNR.

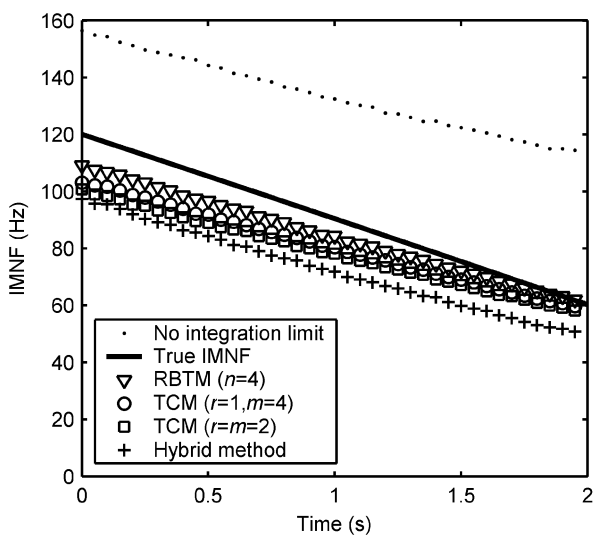

Fig. 4. The mean of 1000 IMNF estimates of a simulated signal with $\mathrm{SNR} = 6\ \mathrm{dB}$ using different methods of estimating the upper integration limit.

When the maximum-frequency estimation methods were used to estimate the integration limit for IMNF estimation, the RBTM gave better results than the TCM, for low SNRs, and much better results than the hybrid method. The mean of the 1000 rms errors can be seen in Fig. 3 for different SNRs. The results show that for SNR below 9 dB the estimation error due to high-frequency noise is of the order of magnitude of the estimation error from TFR estimation or higher. The variance of the rms errors, which could be seen as a stability measure of the methods, was lowest for the RBTM for SNR values lower than 15 dB. For high SNR, the methods performed similarly. As seen in Fig. 4, where the mean of the IMNF estimates obtained with the different integration intervals can be seen for $\mathrm{SNR} = 6\ \mathrm{dB}$, the IMNF estimation error is frequency dependent. The deviation from the true IMNF is larger for higher IMNF for all methods used to determine the integration limit. For high IMNF there is also the largest difference between the methods. The improvement of the RBTM compared to the TCM and the hybrid method is more significant where the IMNF estimation error is large.

VI. Discussion and Conclusion

To limit the integration interval at the high frequencies when calculating IMNF is important when the SNR is low. This paper showed that all tested methods used to determine the integration limit in IMNF estimation improved the estimation of IMNF and that the RBTM outperformed the TCM and the hybrid method.

A way to improve the maximum-frequency estimation is to filter the estimates obtained by the methods investigated in this paper. Filtering the estimates is likely to improve the maximum-frequency estimates, because the maximum frequency is probably, in most cases, smoothly varying.

In all three methods that were compared, there was an assumption of white noise. The noise that always is a part of a real EMG signal is often only approximately white. Baratta *et al.* [2] introduced two methods that can handle colored noise, one which subtracts power line noise and one which subtracts system noise in frequency domain that helps to estimate the median frequency when the SNR is unfavorable. However, the noise spectrum subtraction method uses the power density spectrum calculated with a Fourier transform and is not designed for TFRs (and an extension was not in the scope of this paper).

The detection probability of the methods is a good way to compare their ability of finding the maximum frequency on a theoretical ground. However, if a method is better suited to find a maximum frequency, it will not be a guarantee that the method is better in IMNF estimation, and therefore simulations are important.

The rms errors of the IMNF estimations do not entirely derive from the noise. There are also errors that derive from the filter implementation and from the TFR estimation. These errors are, however, the same for all methods.

As seen in Fig. 4, the IMNF estimation error is frequency dependent. The IMNF values in the simulation were chosen to be as realistic as possible so the rms error should be a measure of overall performance. For extreme IMNF values the methods could probably behave differently.

The Shwedyk model has a relative slow decreasing spectrum. With a faster decrease in the spectrum there is lower signal power in the high frequency region and the noise has more affect on the IMNF estimate. Therefore, it is even more important to limit the integration interval if the signal spectrum has a faster decrease.

In this study, the maximum frequency was used to improve the IMNF estimate. However, the methods could also be applied to higher order spectral moments and to median or instantaneous median frequencies. It is likely that the methods could even be more justified in the estimation of higher order spectral moments because the high-frequency noise has a larger impact on these estimates. The maximum frequency estimated with RBTM could also be used in other applications such as to find the maximum frequency in Doppler signals.

This study adjusted existing maximum-frequency estimation methods to TFRs and introduced a new method that improved the maximum-frequency estimation. The study also confirmed

the importance of limiting the integration interval when estimating IMNF. Based on simulated data, this study showed that IMNF estimates of surface EMG signals containing white noise could be greatly improved for low SNR by applying any of the presented methods and that the RBTM had the best performance of the compared methods.

APPENDIX

Derivation of (17):

Let V be a standard gamma distributed random variable, that is $V \sim G(v|\alpha, 1) = G_V(v|\alpha)$. Suppose that random variables X and Y are related to V through $V = \lambda X = C_L \lambda Y$. Then we have $X \sim G(x|\alpha, \lambda)$, $Y \sim G(y|\alpha, C_L\lambda)$. Now let $T = G^{-1}(P_S|\alpha, \lambda)$. Then $P_S = P(X \leq T) = P(V \leq \lambda T) = G_V(\lambda T|\alpha)$, which implies that $\lambda T = G_V^{-1}(P_S|\alpha)$. Therefore, $G(T|\alpha, C_L\lambda) = P(Y \leq T) = P(V \leq C_L\lambda T) = G_V(C_L\lambda T|\alpha) = G_V(C_L G_V^{-1}(P_S|\alpha)|\alpha)$.

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